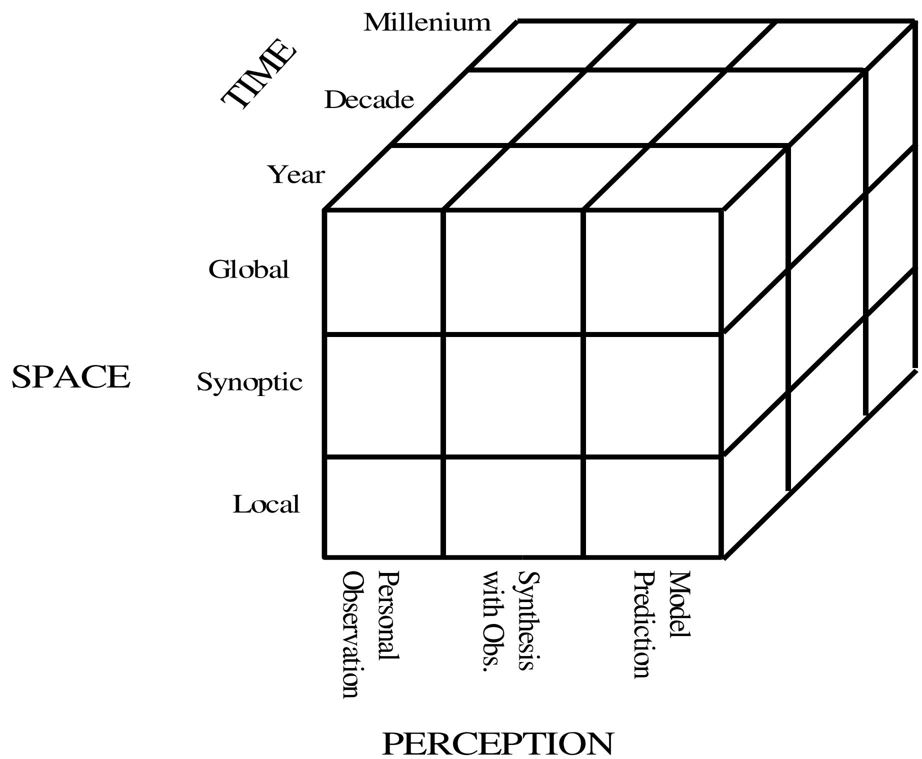
GENERAL CIRCULATION AND CLIMATE

Definitions:

What do we mean by ’General Circulation of the Atmosphere’ and what do we mean by ’Climate’? Both these phrases do not have very unambigious definitions. Depending on whom we ask, their definition may vary somewhat. By ’general circulation of the atmosphere’ we refer to ’the mean condition of the atmosphere’. The word ’mean’ includes a broad range of possibilities. It could refer to time mean circulation. It could also refer to spatial mean circulation such as the zonally averaged circulation. Historically, in the old days of course, all that general circulation meant was the time mean surface circulation. When upper air data became availble, after the World Wars, the atmospheric scientists realized that the middle latitude upper air time mean circulation is amazingly zonal for most of the time. With some waves riding on it. Once in a while these waves amplify and make the circulation non zonal.(show examples). This prompted many scientists to believe that the zonal mean circulation is the fundamental circulation on which the waves and instabilities develop producing the weather systems. Therefore, a lot of efforts were directed to understand the zonally averaged component. However, it was soon realized that the transients and the stationery eddies (or the non zonal component) are indeed important even for the maintainence of the observed zonally averaged component. Thus, it is the totality of the flow with the time mean zonally averaged component supplimented by the eddies that can be considered as the ”general circulation of the atmosphere”. Since , we are interested in the time mean flow, the structure and evolution of individual eddies may not be important. What we shall be interested in are the statistical properties of the eddies. Thus, we shall use the term ′general circulation of the atmosphere′ to refer to basically the three dimensional time mean structure of the atmosphere. This may be characterized by the zonally averaged circulation and the eddy statistics. A bit of ambiguity is still remainig regarding the ″time mean″. We can talk about the monthly mean, the seasonal maean or the annual mean.

Similarly, the word ′Climate′ is used with a wide variety of meanings. To different people, this word has different meaning. For example, to a geologist, the ′climate′ refers to an external agent which forces many of the phenomenon of interest. To an agriculturist, the climate is the ′norm′ upon which the year-to-year or day to day weather is imposed. To most of us, the ′climate′ may be simply mean temperature and rainfall. However, we now know that the temperature and rainfall in any place are manifestations of dynamical balance of the atmospheric fluid over the entire globe. Therfore, a useful definition of the climate may be ″all the statistics of the ocean atmosphere system over an averaged time interval (season, year, decade or longer) computed for the globe or possibly for a selected region″. This definition is still quite broad. Thus, the general circulation of the atmosphere and the climate both refer to the time mean state of the atmosphere and hence are the same. However, in practice when we talk about the general circulation of the atmosphere, we mainly refer to the ″present″ time averaged state of the atmosphere.

On the other hand climate also refers to the variability of the mean state with larger time intervals. Thus, the climate system may be considered to have three axis, space, time and human perception. as shown below in the fig.



The three axes are fundamental, but the divisions are quite arbitary and many more can be included.

Theory of General Circulation

A. Historical Background

Before the world war, the measurement of upper air quantities were almost nonexistent. In olden day's scattered information about surface wind , temperature, pressure and rainfall were collected mainly by sailors as they had to depend on the strength and direction of the wind. There are references to surface wind condition even in very ancient times. For example, A Greek skipper named Hippalus who lived around the time of Christ apparently noted the reversal of the monsoon wind over the Indian Ocean. It seems, he used this knowledge and ventured out ointo the sea instead of hugging close to the coast and reduced the travel time between Egypt to India form 6 months to mere 40 days. Most sailors noted the salient feature of the surface wind and pressure eg:

(1) Surface winds in the tropical region are easterlies while surface winds in the middle latitude are westerlies.

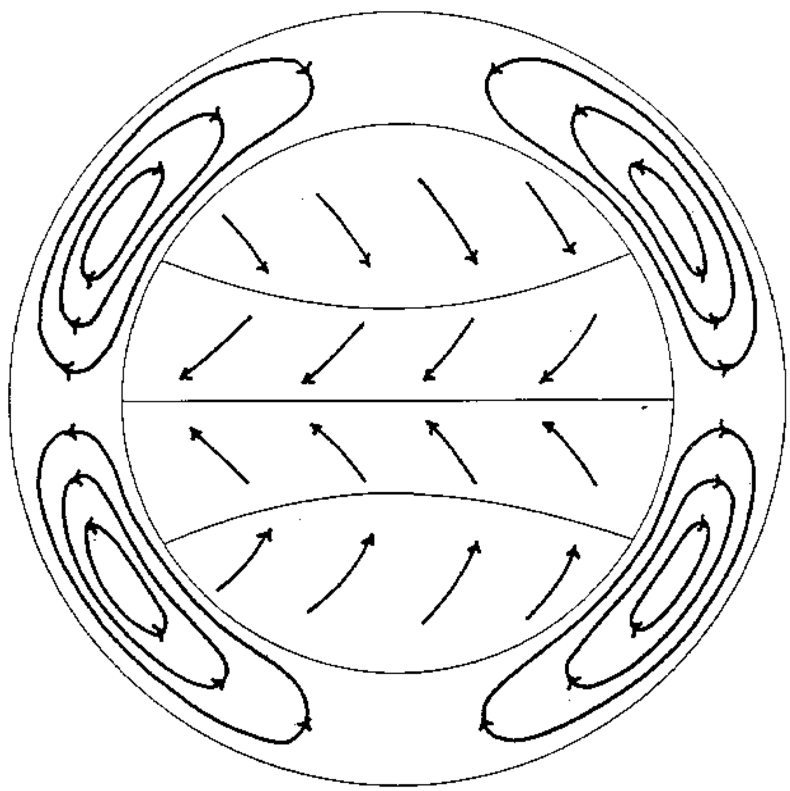
(2) Near the equator, the easterly winds from northern and southern hemisphere seem to converge to a region of low pressure. Now we call this region as the Intertropical convergence zone(ITCZ). The olden day sailors used to find that, the wind in this region converge from north and south and rise. As a result this is a relatively calm region and if their ships got caught in this region, it was difficult to get out. Therefore, they used to call it region of 'doldrums'.

(3) Similarly, the subtropical region between 30° and 45° is a region of high pressure and a region of divergence. Now we know that this is a result of subsidence of air that rises near the tropics (descending limbs of the Hadley circulation). Anyway, this was another region where due to the divergent conditions, winds were very weak and variable. As a result the sailing ships again had hard time getting out of this region. Apparently, many of them used to lighten their ships by throwing out their horses. As a result this region is often refered to as 'horse latitude'.

These rather haphazard knowledge of the surface wind were first consolidated by Edmund Haley in 1686 when he produced charts of average surface wind over the ocean. His charts truned out to be tremendous time saver for many sailors and of immense commercial value. He also tried to give a common explanation for the north easterly and south easterly trades. For the first time, he recognised that air *converges* to the *hottest* region. He recognised that the equatorial region is the hottest region and the north easterly trades in the north and south easterly trades in the south are results of air converging to this hottest region. He explained the seasonal migration of the trades as a result of the seasonal migration of the hottest belt. Well, this is all nice, but why don't the winds blow directly from north or south? Haley did not have a good explanation for this. For some unclear reasons, he seemed to think that due to the diurnal variation of the solar heating, the winds will be more strongly veered toward the evening sun than toward the morning sun so that we shall see an easterly wind on the average. Although Haley recognised the importance of air rising over heated region, he did not recognise the law of continuity of mass and hence could not predict the meridional cells. He also had no idea about the angular momentum conservation or the deflecting effect of the earths rotation.

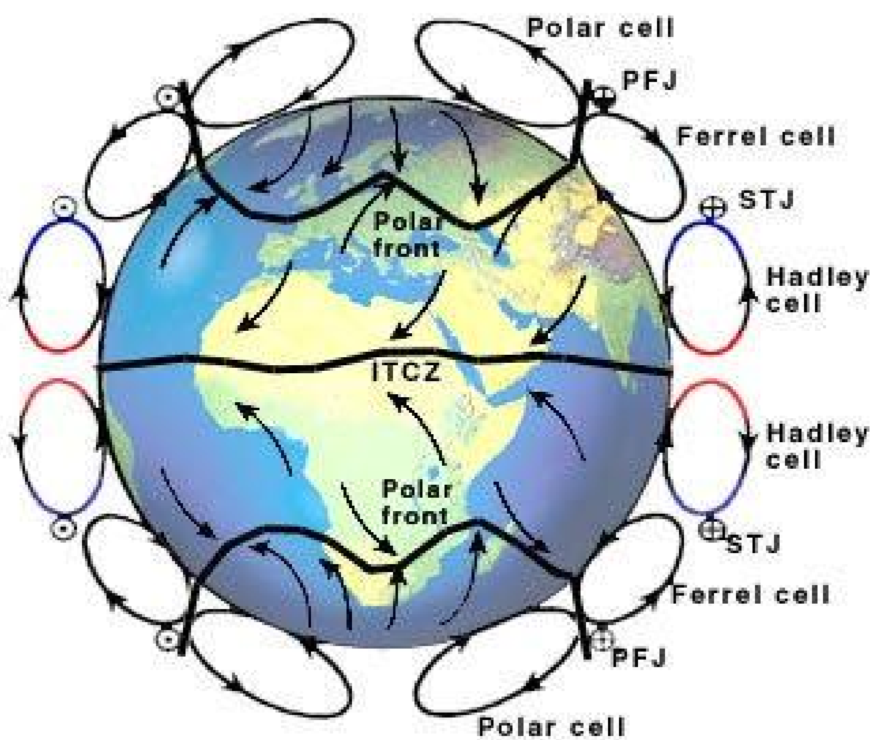
In 1735, George Hadley provided a better explanation for the equatorial trade winds. He rejected the idea that motion towards the sun would result in any average westward or eastward movement. He concluded that the distribution of solar heating would lead to a general rising motion in the lower latitude and general sucking motion in the higher latitude. This circuit will be completed by equator ward motion in the lower levels and poleward motion in the upper levels. He somehow recognised the need for continuity of the motion. He also recognised the role of the earths rotation. He noted that due to the earths rotation, the earths surface moves most rapidly eastward in the lowest latitudes. So, air moving initially equatorward with no relative eastward or westward motion would eventually acquire a westward velocity relative to the earth. This he concluded because he thought the air will conserve its absolute velocity. He realized that if an air parcel moves this way over a long distance it would acquire large easterly velocity. Then he proposed that frictional drag of the earth surface will reduce the velocities to their observed values. He, then recognised that over a long period of time, the easterly momentum gained by the solid earth from the descelaration of the easterly wind will slow down the earth unless there is compensating amount of westerly momentum at some other latitude. He concluded that the westerly surface winds in the middle latitude provided this gain in the westerly momentum. But what produces the westerly wind? Hadley suggested that the air that rise over the low latitude and move toward high latitudes will acquire strong westerly relative velocity when they reach high latitudes where they sink and flow toward equatorward. When they reach the lower levels they would still have westerly momentum. So, this equatorward flow would initially be westerly. As it moves toward equatorward, it would aquire easterly momentum and reduce the westerly velocities to zero at some latitude and start becoming easterly equatorward of that latitude.

A schematic picture of the Hadley's model of the general circulation is shown below

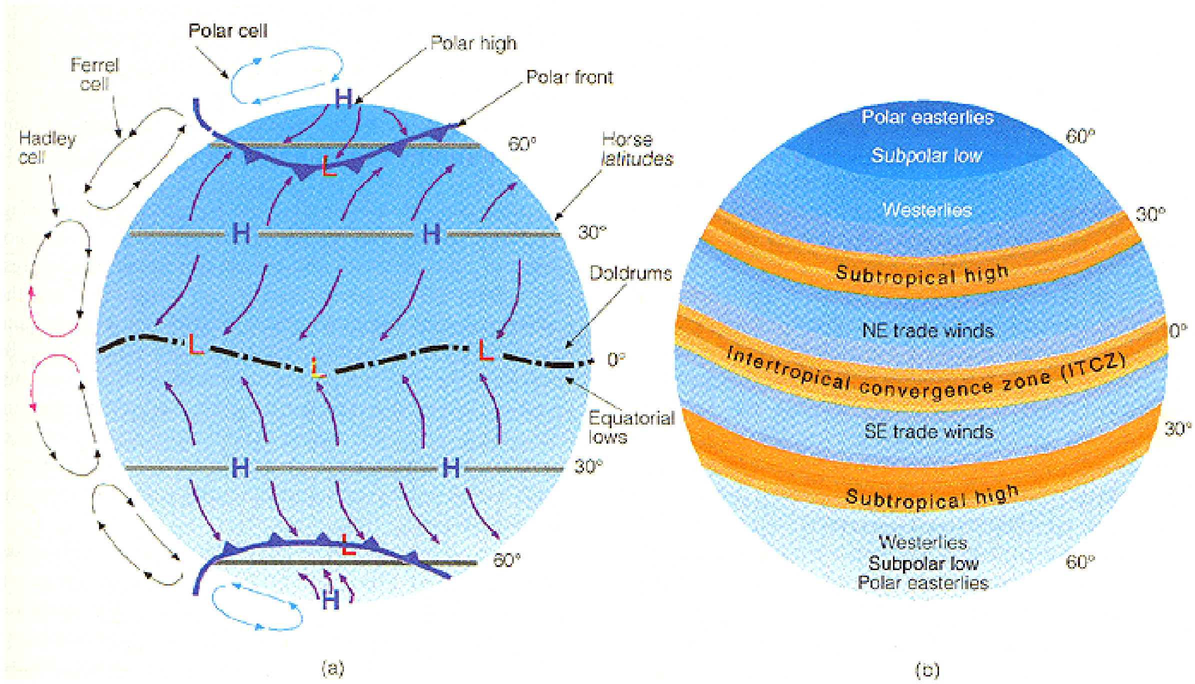


Hadley's insight into the general circulation at a time when he wrote this paper is quite remarkable. At the time the existance of the polar easterlies were not known and thus his theory seem to have explained most of the observed balance requirements. Although he wrongly assumed that the *absolute velocity* rather than the *absolute angular momentum* was conserved, it was remarkable that he recognised the right role of the earths rotation. It took nearly one century before Coriolis correctly formulated the deflecting effect of the earths rotation. One important drawback of the Hadley's theory became apparent in ninteenth century when more observations became available. These observations showed the westerlies in higher latitudes tended to blow from some what south of west rather than north of west as demanded by Hadley's theory!

After Hadley's work for more than half a century the scientific community was generally unaware of its existence. No significant development on the theory of the general circulation also took place during this period. In the ninteenth century, Dove (1837) and Maury(1855) introduced some new ideas. But the real important contribution came from William Ferrel in 1856. The circulation envisioned by Ferrel is schematically shown in Fig.2 below.



Ferrel agreed with Hadley that the main moving force for driving the



atmospheric motion is the pole to equator density gradient created by the solar heating. This leads to meridional motions and through the action of the east-west Coriolis force would result in easterly and westerly surface wind. However, he noted that Hadley's model could not explain the high pressure zone over the horse latitudes as well as the south westerlies north of the horse latitudes. In order to explain these factors he proposed that the general circutlation of the atmosphere consists of not one but three cells. One direct cell in the tropics just as Hadley thought, one indirect cell in the middle latitudes and another direct cell in the polar latitudes. He claimed that the descending air from the two cells over the horse latitudes will create the high pressure region over there. The greatest contribution of Ferrel was the correct formulation of the Coriolis force. With this he was able to explain correctly the north easterlies in the tropics and southwesterlies in the middle latitude. He was also able to formulate the geostrophic relations and thermal wind relations. But he had no clear idea about the origin of the indirect cell in the middle latitude. He actually tried to formulate mathematical equations governing the general circulation and tried to get solution of these equations. However he felt that this could not be done because, the frictional forces could not be formulated properly. Ferrel updated this theory of the general circulation in 1859 and again in 1889.

It was Oberbeck(1888) who tackled the problem that Ferrel thought was unfeasible. Oberbeck represented the friction by a simple coefficient of viscosity and tried to get some analytic solution of the general circulation. Like Ferrel he sought to derive motion from the temperature field and did not have a thermodynamic equation. He represented the temperature field by a simple analytic function of latitude and elevation. He first sought the circulation that would exist in the absence of rotation and advection. Next he tried to get solutions representing balance between Coriolis force and friction. Oberbeck's solutions were far from realistic and he could not get solutions under realistic approximations due to mathematical difficulties. Nevertheless, these attempts at deriving analytic solutions of the circulation were pioneering in their own right.

Untill the begining of the twentieth century, however, all the studies assumed that the general circulation is perfectly symmetric. In the early twentieth century, scientists started realising the importance of the cyclones and storms or the large scale eddies in the general circulation. It was clear that in practice the circulation is not quite zonally symmetric. Then the questions arose. (1) what is responsible for the origin of these cyclones and storms (2) even though the general circulation is not quite zonally symmetric due to the pressure of these large scale eddies, if we imagine a zonally averaged circulation, what role these large scale eddies would play in maintaining this hypothetical zonally averaged circulation.

It was A. Defant (1921) who for the first time showed that the large scale eddies could accomplish a large transport of heat across latitude circles. Defant introduced that idea that the motion in the middle latitude atmosphere is simply turbulance in a very large scale. The individual cyclones and anticyclones can be considered as individual turbulent elements. As in the fluid turbulance, he proposed that these large eddies could transport necessary amount of heat form low to high latitudes. In fact he went a step further and tried to calculate this transport by applying the recently formulated mixing length theory of trubulance. He assumed a mixing length of about 15° latitude and an average north south wind component of about $3ms^{-1}$ and calculated a horizontal Austausch coefficient (ratio of the eddy transport to the gradient of sensible heat) of about $5 \times 10^7 gcm^{-1}s^{-1}$. This is about a million times larger than the Austausch coefficient that would be expected from small scale turbulance theory. Choosing different values of the Austausch coefficient between 5×10^7 and 5×10^8, he calculated the temperature distribution that should prevail north of 30°N. Using value of the Austausch coefficient of the order of 10^8, he found that calculated temperature distribution agreed fairly well with the observations. This, for the first time demonstrated the importance of the large scale eddies in maintaining the observed zonally averaged circulation by providing part of the heat transport.

However, the very existance of these cyclones and storms was not explained. In 1937, V. Bjerknes proposed that a symmetric general circulation in principle is possible , but such a symmetric circulation is unstable with respect to small symmetric perturbation. He said that the circulation that would prevail in the absence of depertures from zonal symmetry was essentially the one proposed by Ferrel and Thompson with a large direct cell in either hemisphere with a shallow indirect cell in the low levels in middle latitudes. This instability of the circulation will result in the fully developed cyclones and anticyclones and make the circulation asymetric. This description is closer to our present understanding of the general circulation which we shall discuss in detail little later in our course.